

DISCRETE RANDOM VARIABLE/DISCRETE PROBABILITY DISTRIBUTION

Handout #2 P&S (MT-331)

DISCRETE RANDOM VARIABLE

1. The probability distribution of the discrete random variable X is Find the mean of X.

$$f(x) = \binom{3}{x} \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{3-x}, \quad x = 0, 1, 2, 3.$$

2. Roulette wheel is divided in to 6 sectors of unequal area marked with the no's 1, 2, 3, 4, 5, & 6. The wheel is spun & X is the random variable the number on which the wheel stops. The probability of X is as follows:

x	1	2	3	4	5	6
P(X=x)	1/16	3/16	k	1/4	3/16	1/16

Find the value of 'k' and find (i) $E(3x-5)$ (ii) $E(6x^2 + 6x - 10)$

3. Suppose that an antique jewelry dealer is interested in purchasing a gold necklace for which the probabilities are 0.22, 0.36, 0.28, and 0.14, respectively, that she will be able to sell it for a profit of \$250, sell it for a profit of \$150, break even, or sell it for a loss of \$150. What is her expected profit?
4. An attendant at a car wash is paid according to the number of cars that pass through. Suppose the probabilities are 1/12, 1/12, 1/4, 1/4, 1/6, and 1/6, respectively, that the attendant receives \$7, \$9, \$11, \$13, \$15, or \$17 between 4:00 P.M. and 5:00 P.M. on any sunny Friday. Find the attendant's expected earnings for this particular period.
5. The number of messages sent per hour over a computer network has the following distribution:

$x = \text{number of messages}$	10	11	12	13	14	15
$f(x)$	0.08	0.15	0.30	0.20	0.20	0.07

Determine the mean and standard deviation of the number of messages sent per hour.

6. The monthly demand for transistors is known to have the following probability distribution.

Demand (n)	1	2	3	4	5	6
Probability (P)	0.10	0.15	0.20	0.25	0.18	0.12

Determine the expected demand for transistors. Also obtain the variance?

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SE (EE) SPRING SEMESTER 2024
CHAPTER 4 & 5
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BINOMIAL & MULTINOMIAL DISTRIBUTION

- 1 Twelve people are given two identical speakers, which they are asked to listen to for differences, if any. Suppose that these people answer simply by guessing. Find the probability that three people claim to have heard a difference between the two speakers.
- 2 In a certain city district, the need for money to buy drugs is stated as the reason for 75% of all thefts. Find the probability that among the next 5 theft cases reported in this district,
 - (a) Exactly 2 resulted from the need for money to buy drugs?
 - (b) At most 3 resulted from the need for money to buy drugs?
- 3 2% of the batteries produced by a factory are not satisfactory. In a sample of 12 batteries chosen at random, find the probability that:
 - a) Exactly 3 batteries are not satisfactory?
 - b) Less than 2 batteries are not satisfactory?
- 4 According to a survey by the Administrative Management Society, one-half of U.S. companies give employees 4 weeks of vacation after they have been with the company for 15 years. Find the probability that among 6 companies surveyed at random, the number that give employees 4 weeks of vacation after 15 years of employment is
 - (a) anywhere from 2 to 5;
 - (b) Fewer than 3.
- 5 One prominent physician claims that 70% of those with lung cancer are chain smokers. If his assertion is correct,
 - (a) find the probability that of 10 such patients recently admitted to a hospital, fewer than half are chain smokers;
 - (b) Find the probability that of 20 such patients recently admitted to a hospital, fewer than half are chain smokers.
- 6 In testing a certain kind of truck tire over rugged terrain, it is found that 25% of the trucks fail to complete the test run without a blowout. Of the next 15 trucks tested, find the probability that
 - (a) from 3 to 6 have blowouts;
 - (b) fewer than 4 have blowouts;
 - (c) More than 5 have blowouts.
- 7 The probability that a patient recovers from a delicate heart operation is 0.9. What is the probability that exactly 5 of the next 7 patients having this operation survive?
- 8 A traffic control engineer reports that 75% of the vehicles passing through a checkpoint are from within the state. What is the probability that fewer than 4 of the next 9 vehicles are from out of state?
- 9 It is known that 60% of mice inoculated with a serum are protected from a certain disease. If 5 mice are inoculated, find the probability that
 - (a) none contracts the disease.
 - (b) fewer than 2 contract the disease.
 - (c) More than 3 contract the disease.
- 10 Suppose that airplane engines operate independently and fail with probability equal to 0.4. Assuming that a plane makes a safe flight if at least one-half of its engines run, determine whether a 4-engine plane or a 2-engine plane has the higher probability for a successful flight.

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- 11 A multiple choice test contains 25 questions, each with four answers. Assume a student just guesses on each question.
(a) What is the probability that the student answers more than 20 questions correctly?
(b) What is the probability the student answers less than 5 questions correctly?
- 12 The probabilities are 0.4, 0.2, 0.3, and 0.1, respectively, that a delegate to a certain convention arrived by air, bus, automobile, or train. What is the probability that among 9 delegates randomly selected at this convention, 3 arrived by air, 3 arrived by bus, 1 arrived by automobile, and 2 arrived by train?
- 13 A company generally purchases large lots of a certain kind of electronic device. A method is used that rejects a lot if 2 or more defective units are found in a random sample of 25 units.
(a) What is the probability of rejecting a lot that is 1% defective?
(b) What is the probability of accepting a lot that is 5% defective?
- 14 The probability of a bomb hitting a target is 0.20. Two bombs are enough to destroy a bridge. If 6 bombs are aimed at the bridge, find the probability that the bridge will be destroyed.
- 15 A safety engineer claims that only 40% of all workers wear safety helmets when they eat lunch at the workplace. Assuming that this claim is right, find the probability that 4 of 6 workers randomly chosen will be wearing their helmets while having lunch at the workplace.

NEGATIVE BINOMIAL DISTRIBUTION , GEOMETRIC DISTRIBUTION & POISSON DISTRIBUTION

- 16 The probability that a person living in a certain city owns a dog is estimated to be 0.3. Find the probability that the tenth person randomly interviewed in that city is the fifth one to own a dog.
- 17 Find the probability that a person flipping a coin gets
(a) the third head on the seventh flip;
(b) The first head on the fourth flip.
- 18 Three people toss a fair coin and the odd one pays for coffee. If the coins all turn up the same, they are tossed again. Find the probability that fewer than 4 tosses are needed.
- 19 A scientist inoculates mice, one at a time, with a disease germ until he finds 2 that have contracted the disease. If the probability of contracting the disease is $1/6$, what is the probability that 8 mice are required?
- 20 The probability that a student pilot passes the written test for a private pilot's license is 0.7. Find the probability that a given student will pass the test
(a) on the third try.
(b) Before the fourth try.
- 21 Suppose the probability that any given person will believe a tale about the transgressions of a famous actress is 0.8. What is the probability that
(a) The sixth person to hear this tale is the fourth one to believe it?
(b) The third person to hear this tale is the first one to believe it?

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- 22 An inventory study determines that, on average, demands for a particular item at a warehouse are made 5 times per day. What is the probability that on a given day this item is requested
(a) More than 5 times?
(b) Not at all?
- 23 A certain area of the eastern United States is, on average, hit by 6 hurricanes a year. Find the probability that in a given year that area will be hit by
(a) fewer than 4 hurricanes.
(b) Anywhere from 6 to 8 hurricanes.
- 24 On average, 3 traffic accidents per month occur at a certain intersection. What is the probability that in any given month at this intersection
(a) Exactly 5 accidents will occur?
(b) Fewer than 3 accidents will occur?
(c) At least 2 accidents will occur?
- 25 A restaurant chef prepares a tossed salad containing, on average, 5 vegetables. Find the probability that the salad contains more than 5 vegetables
(a) on a given day
(b) on 3 of the next 4 days
(c) For the first time in April on April 5.
- 26 The probability of a successful optical alignment in the assembly of an optical data storage product is 0.8. Assume the trials are independent.
(a) What is the probability that the first successful alignment requires exactly four trials?
(b) What is the probability that the first successful alignment requires at most four trials?
(c) What is the probability that the first successful alignment requires at least four trials?
- 27 The number of cracks in a section of interstate highway that are significant enough to require repair is assumed to follow a Poisson distribution with a mean of two cracks per mile.
(a) What is the probability that there are no cracks that require repair in 5 miles of highway?
(b) What is the probability that at least one crack requires repair in mile of highway?
- 28 The probability that a person will die from a certain respiratory infection is 0.002. Find the probability that fewer than 5 of the next 2000 so infected will die.
- 29 Suppose a life insurance company insures the lives of 5000 persons aged 42. If studies show the probability that any 42-years old person will die in a given year to be 0.001, find the probability that the company will have to pay at least two claims during a given year.
- 30 Suppose that, on average, 1 person in 1000 makes a numerical error in preparing his or her income tax return. If 10,000 forms are selected at random and examined, find the probability that 6, 7, or 8 of the forms contain an error.

HYPERGEOMETRIC & MULTIVARIATE HYPERGEOMETRIC DISTRIBUTION

- 31 A homeowner plants 6 bulbs selected at random from a box containing 5 tulip bulbs and 4 daffodil bulbs. What is the probability that he planted 2 daffodil bulbs and 4 tulip bulbs?
- 32 To avoid detection at customs, a traveler places 6 narcotic tablets in a bottle containing 9 vitamin tablets that are similar in appearance. If the customs official selects 3 of the tablets at random for analysis, what is the probability that the traveler will be arrested for illegal possession of narcotics?

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- 33 From a lot of 10 missiles, 4 are selected at random and fired. If the lot contains 3 defective missiles that will not fire, what is the probability that
(a) All 4 will fire?
(b) At most 2 will not fire?
- 34 What is the probability that a waitress will refuse to serve alcoholic beverages to only 2 minors if she randomly checks the IDs of 5 among 9 students, 4 of whom are minors?
- 35 A company is interested in evaluating its current inspection procedure for shipments of 50 identical items. The procedure is to take a sample of 5 and pass the shipment if no more than 2 are found to be defective. What proportion of shipments with 20% defectives will be accepted?
- 36 A foreign student club lists as its members 2 Canadians, 3 Japanese, 5 Italians, and 2 Germans. If a committee of 4 is selected at random, find the probability that
(a) All nationalities are represented.
(b) All nationalities except Italian are represented.
- 37 An urn contains 3 green balls, 2 blue balls, and 4 red balls. In a random sample of 5 balls, find the probability that both blue balls and at least 1 red ball are selected.